

Prabhdeep Singh Sethi

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Computer Vision (Robotics Institute, School of Computer Science) (GPA: 4.04/4)

Dec 2024

Coursework: Advanced Computer Vision, 3D Vision, Multimodal Learning, Reinforcement Learning, Geometric Methods in Vision

Government College of Engineering, Nagpur

Nagpur, India

Bachelor of Engineering in Computer Science (GPA: 9.5/10)

Aug 2021

Coursework: Operating Systems, Data Structures & Algorithms, Artificial Intelligence, Databases, Object Oriented Programming

EXPERIENCE

Snap Inc

Los Angeles, CA

Machine Learning Engineer, Generative 3D Team

Feb 2025 - Current

- Led E2E development (R&D → production) of a text/image-driven [3D Gaussian Splatting pipeline](#) serving 3B+ impressions.
- Achieved 40× speedup (600s → 15s) by replacing legacy optimization-based generation with a feedforward approach.
- Integrated generative image models (Flux, QIE) and improved multi-view geometry via video-to-video diffusion model.
- Contributed to the earlier optimization-based pipeline, enhancing asset fidelity at scale for systems serving 10B+ impressions.

Apple

Sunnyvale, CA

Computer Vision Research Intern, Vision Pro Team

May 2024 - Aug 2024

- Built an automated pipeline for immersive environment creation using 3D reconstruction for AR/VR applications.
- Accelerated SfM on Apple Metal (+5%) and improved 3D Gaussian Splatting reconstruction quality (+26% PSNR).

Wobot Intelligence

New Delhi, India

Computer Vision Engineer-II

Feb 2022 - Aug 2023

- Led a team of 6 to deliver vehicle and person re-identification solutions, serving 1M+ cameras and 10K+ customers.
- Reduced false IDs by 65% & improved Rank-1 by 35% in multi-camera tracking using improved features for identity matching

Solar Industries India Ltd. (Research and Development Lab)

Nagpur, India

Computer Vision Researcher

Aug 2021 - Jan 2022

- Designed a pipeline for inspection of critical military parts using a Vision Transformer achieving 96.5% mAP across 9 parts.
- Developed real-time overspeeding detection using YOLOv4, PaddleOCR, and DeepSORT, deployed via NVIDIA Triton.

PUBLICATIONS

S Jain*, A Kuthiala*, **PS Sethi**, P Saxena, “StyleSplat: 3D Object Style Transfer with Gaussian Splatting”

Oct 2024

arXiv, 2024 [\[Project Page\]](#)

- Developed a localized 3D style transfer method for Gaussian Splatting with joint object segmentation and stylization.
- Enabled controllable multi-object stylization via view-consistent Gaussian selection and nearest-neighbor feature matching.

PS Sethi*, A Agrawal*, CMS Lezcano*, I Heredia*, “Listen Then See: Video Alignment with Speaker Attention”

Feb 2024

Proceedings of CVPR Workshops, 2024 [\[Project Page\]](#)

- Addressed VLM bias due to reliance on questions by designing a multimodal fusion strategy leveraging audio, video & text.
- Improved modality utilization (+5%) and achieved +3.5% performance gain on Social IQ 2.0.

R Zawar, **PS Sethi**, R Roy, “Jensen-Shannon Divergence in Safe Multi-Agent RL”

Dec 2023

ICLR, Tiny Paper Track, 2024 [\[Paper\]](#)

- Proposed a safer alternative in MACPO with JS divergence, achieving higher rewards and reduced costs in multi-agent systems.

RESEARCH PROJECTS

Generalizable Sparse view 3D Object Reconstruction (3D Gaussian Splatting) [\[Project Page\]](#)

Jan 2024 - Dec 2024

Advisor: [Dr. Shubham Tulsiani](#), Physical Perception Lab

- Developed a feed forward network to predict 3D Gaussian attributes for objects in sparse (4-view) and unposed settings.
- Proposed a feedforward, non-pixel-aligned approach leveraging transformer-based representations for generalizable 3D.

GIF Tune (Video Generation) [\[Project Page\]](#)

Feb 2024 - Apr 2024

- Developed GIF-Tune, a one-shot tuning strategy enabling continuous text-to-GIF synthesis from text prompts.
- Enhanced the output using depth-conditioned Stable Diffusion and 3D temporal attention layers from a single text-GIF pair.

SKILLS

Languages and Frameworks: Python, C++, Swift, Bash, SQL; PyTorch, PyTorch3D, TensorFlow, Numpy, OpenCV, Flask

Tools and Platforms: Docker, Kubernetes, Triton, Git, DeepStream, Ray, TensorRT, AIMET, AWS, Azure